

ARGENTINE ASSOCIATION FOR INDUSTRIAL, COMPUTATIONAL AND APPLIED MATHEMATICS. MODEL ARTICLE

Alan Matys^b and Federico Rodriguez[†]

^bUniversidad del CEMA, alan.matys92@gmail.com, federico.m.rodriguez.h@gmail.com

Abstract: This article is intended to tackle the portfolio-building problem by framing it as a budget allocation challenge, balancing expected returns and risk. Traditional methods like the Critical Line Algorithm (CLA) face diversification limitations. To address this, Hierarchical Risk Parity (HRP) offers a more structured approach, clustering correlated assets and assigning weights top-down. By leveraging the covariance matrix, HRP enhances diversification and aligns with real-world portfolio management practices. This methodology will be applied in the cryptocurrency domain to optimize asset allocation and risk management.

Keywords: *Hierarchical Risk Parity, Risk-based asset allocation, Cryptocurrency, Portfolio Optimization*
2000 AMS Subject Classification: 21A54 - 55P54

1 INTRODUCTION

The portfolio construction problem can be framed as a budget allocation challenge, where each point in the allocation space represents a portfolio with different asset weightings. This problem is traditionally modeled in terms of **expected return** and **risk**, understood as the deviation of asset returns.

In a **Capital Asset Pricing Model (CAPM)** framework, portfolio optimization follows two main objectives:

- Given a fixed risk level, **maximize return**.
- Given a target return, **minimize associated risk**.

For each portfolio, the **portfolio return** is a random variable, composed of the weighted sum of individual asset returns:

$$R_p = \sum_{i=1}^N W_i R_{A_i}$$

where:

- R_p : Portfolio return
- N : Total number of assets in the portfolio
- A_i : Asset i
- R_{A_i} : Return of asset A_i
- W_i : Budget weight assigned to asset A_i , with $\sum_{i=1}^N W_i = 1$.

A key challenge in this approach is the dependence on **unobservable parameters**. The **mean and variance** of asset returns are population characteristics, which are unknown and can only be estimated from historical data within a given time window.

One of the traditional methods for portfolio optimization is the **Markowitz Critical Line Algorithm (CLA)**, which solves the allocation problem by minimizing variance subject to return estimates over a defined time window. While the algorithm provides a theoretically optimal solution, it has both advantages and disadvantages:

- **Pros:** Ensures weight convergence after a finite number of steps.
- **Cons:** Prone to instability, high concentration in specific assets, and potential underperformance.

Moreover, CLA requires inverting the variance-covariance matrix, a process that can be straightforward but, depending on the matrix structure, may lead to numerical instability and unreliable results.

Risk-based asset allocation methods address some of the challenges faced by the **CLA**. The goal is to diversify the portfolio based on information contained in the **covariance matrix**. Among these approaches, **Hierarchical Risk Parity (HRP)** introduces a novel weighting strategy:

This article explores the application of **HRP in the cryptocurrency domain**, leveraging its ability to enhance diversification while addressing estimation uncertainty in risk and return parameters.

2 HRP PORTFOLIO BUILDING

The **Hierarchical Risk Parity (HRP)** approach aims to diversify the portfolio based on information contained in the **covariance matrix**. Unlike traditional optimization methods, HRP introduces a hierarchical allocation structure that offers several advantages:

- Weights are assigned hierarchically, grouping highly correlated assets before distributing weights within each group.
- Similar assets receive allocations first, followed by further subdivisions.
- Weights are assigned in a **top-down** manner, mirroring how many portfolio managers construct their portfolios.

A key advantage of HRP is its ability to function even when the covariance matrix is singular or lacks a well-defined inverse. This is achieved through a structured three-step process:

2.1 HIERARCHICAL CLUSTERING

The first step involves identifying clusters of assets based on their **correlation matrix**. By grouping highly correlated assets together, HRP constructs a hierarchical representation of the asset relationships.

2.2 QUASI-DIAGONALIZATION

Once clusters are identified, the next step is to reorder them based on their **risk profile**. This process ensures that assets with distinct risk characteristics are positioned effectively within the portfolio structure.

2.3 RECURSIVE BISECTION

The final step involves a **recursive bisection** process, where weights are assigned in a disaggregated manner. Instead of directly optimizing the covariance matrix inversion, HRP distributes weights progressively across the hierarchy, ensuring a stable and diversified allocation.

2.4 HIERARCHICAL CLUSTERING

The first step in HRP is the construction of a hierarchical clustering structure based on asset correlations. This process begins by computing a **distance matrix** derived from the correlations of N assets.

The correlation matrix is defined as:

$$\rho = (\rho_{i,j})_{i,j=1,\dots,N}$$

where each element $\rho_{i,j}$ represents the correlation between asset X_i and asset X_j :

$$\rho_{i,j} = \rho[X_i, X_j].$$

Since the correlation matrix is bounded within $[-1, 1]$, its scale does not affect the clustering process.

From this correlation matrix, we construct a **distance metric** to measure the dissimilarity between asset pairs. The metric is defined as:

$$d : (X_i, X_j) \subseteq \mathcal{B} \rightarrow \mathbb{R} \in [0, 1], \quad d_{i,j} = d[X_i, X_j] = \sqrt{\frac{1}{2}(1 - \rho_{i,j})}.$$

where $\mathcal{B}$ represents the Cartesian product of asset pairs.

This transformation ensures that assets with high correlations are placed closer together in the hierarchical structure, forming the basis for further steps in the HRP allocation process.

With the previously computed distance matrix, we calculate the **Euclidean distances** between the columns of the matrix:

$$\tilde{d} : (D_i, D_j) \subseteq \mathcal{B} \rightarrow \mathbb{R} \in [0, \sqrt{N}], \quad \tilde{d}_{i,j} = \tilde{d}[D_i, D_j] = \sqrt{\sum_{n=1}^N (d_{n,i} - d_{n,j})^2}.$$

This results in $\tilde{d}_{i,j}$ being a function of the entire correlation matrix, rather than just an individual correlation pair.

Finally, assets are grouped based on the final matrix using a **linkage criterion**, determining the hierarchical structure necessary for the next steps of the HRP process.

2.5 QUASI-DIAGONALIZATION

In this step, the matrix is reorganized such that rows and columns are arranged so that the largest values in the distance matrix are grouped along the diagonal.

Based on the clusters identified in the previous step, we reorder the matrix so that similar assets are positioned closer together, while uncorrelated or negatively correlated assets are placed farther apart. This structured ordering allows for a clearer representation of asset relationships and improves the stability of subsequent calculations.

As we move away from an asset's cluster, correlations decrease, reinforcing the separation of distinct investment groups. By achieving a **quasi-diagonal structure**, this step significantly enhances the efficiency of further computations, particularly in weight allocation and portfolio construction.

2.6 RECURSIVE BISECTION

Using the **quasi-diagonalization** from the previous step, we now determine the **optimal portfolio allocation** by applying the **Inverse Variance Portfolio (IVP)** method within each group.

The IVP approach assigns portfolio weights inversely proportional to the variance of each asset, ensuring a risk-balanced allocation. The recursive bisection process follows these steps:

- First, allocate weights to each major group identified in the clustering step.
- Then, recursively divide the assigned weights within each subgroup.
- Continue subdividing until further partitioning is no longer possible.

By **iteratively splitting and distributing weights**, this method ensures a stable and diversified allocation, avoiding over-concentration in specific assets. The recursive approach allows HRP to dynamically adjust portfolio weights while maintaining an optimal risk structure.

3 RESULTS AND EVALUATION METHODOLOGY

The proposed method will be tested and compared against other portfolio allocation techniques, namely the **Inverse Variance Portfolio (IVP)** and the **Minimum Variance Portfolio (MVP)**. In addition to

these benchmarks, the strategy will also be evaluated against simple **BTC holding** and **ETH holding** for the entire backtesting period.

For this analysis, the backtesting period spans **two years**, from **January 1, 2021, to December 31, 2023**.

3.1 COMPARISON SCENARIOS

The methodology will be assessed under two distinct scenarios:

- **Static Portfolio Allocation:** Based on the initial data, portfolio weights will be assigned once at the beginning of the backtesting period, with no reweighting throughout the entire duration.
- **Weekly Rebalancing:** The best-performing method from the static allocation scenario will be compared against the three methodologies under a **weekly rebalancing** strategy.

3.2 EVALUATION METRICS

To evaluate the performance of each portfolio strategy, the following metrics will be computed:

- **Total Return (%)**: The overall percentage gain or loss over the backtesting period.
- **Annualized Volatility (%)**: Measures the standard deviation of returns on an annual basis.
- **Sharpe Ratio**: Evaluates risk-adjusted returns by comparing excess returns over the risk-free rate relative to volatility.
- **Sortino Ratio**: Similar to the Sharpe Ratio but only considers downside risk.
- **Maximum Drawdown (%)**: The largest peak-to-trough decline in portfolio value during the period.
- **Calmar Ratio**: The return-to-risk measure comparing total return to maximum drawdown.

These metrics will provide a comprehensive assessment of the risk-adjusted performance and robustness of the HRP strategy in comparison to other allocation methods.

3.3 FINAL RESULTS

In this section, we present the results for both portfolio allocation scenarios, focusing on **Cumulative Returns** and **Metrics Evaluation**.

3.3.1 Static Portfolio Allocation Results

3.3.2 Weekly Rebalancing Results

4 CONCLUSION & FUTURE RESEARCH

4.1 CONCLUSION

Diversificación: In the case of portfolios without rebalancing, the capital allocation is more evenly distributed. However, for the scenario with rebalancing, the opposite occurs, leading to more concentrated allocations over time.

Rebalanceo: In the MVP strategy, each rebalancing step can lead to significantly different asset allocations. This results in high transaction costs, which reduce net returns.

4.2 FUTURE RESEARCH

Denosing: Addressing noise caused by very small eigenvalues in the correlation matrix to improve estimation stability.

Denoting: Adjusting for correlation shocks by filtering out market-wide movements where all assets move in the same direction.

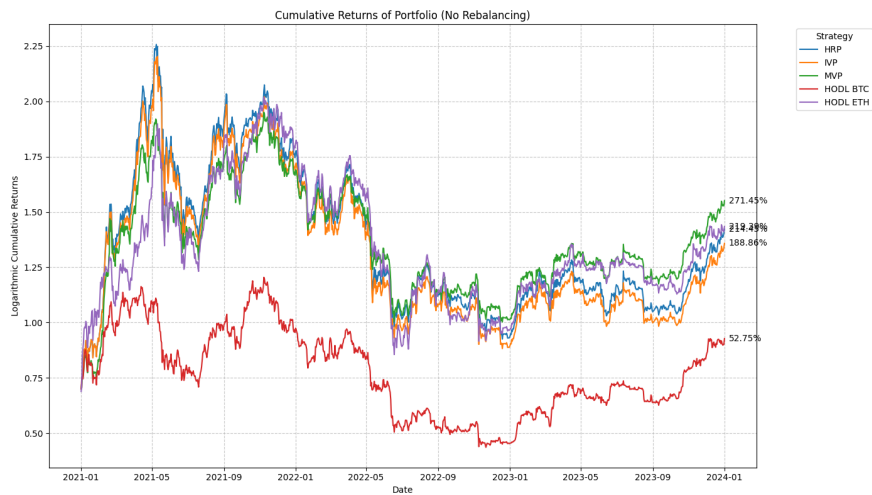


Figure 1: Cumulative Returns for Static Portfolio Allocation.

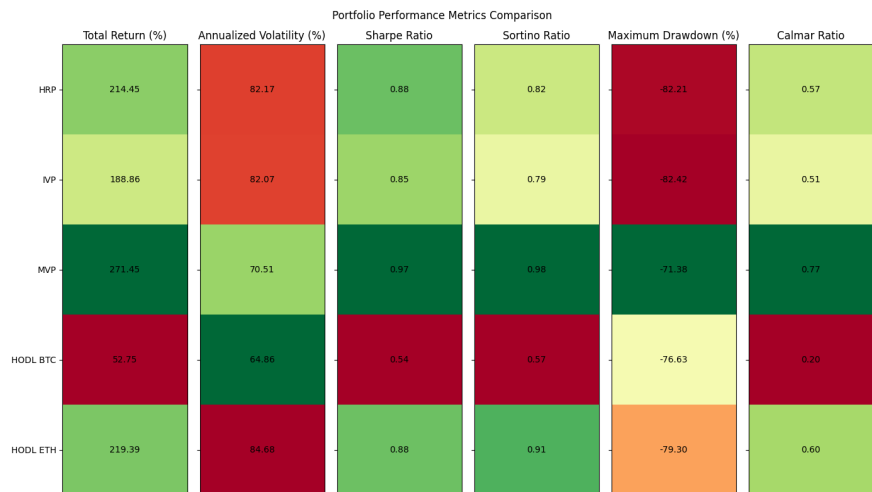


Figure 2: Metrics Evaluation for Static Portfolio Allocation.

ACKNOWLEDGEMENTS

We want to thank all the ****Ucema Quant Course professors and staff****, especially ****Manuel Maurette, Emiliano Delefa, and Pablo Martín Gechidijian****. Finally, we extend our gratitude to ****Marcos López de Prado**** for all the knowledge shared in his bibliography on this subject.

REFERENCES

REFERENCES

- [1] López de Prado, M. (2018). *Advances in Financial Machine Learning*. John Wiley & Sons.
- [2] López de Prado, M. (2020). *Machine Learning for Asset Managers*. Cambridge University Press.

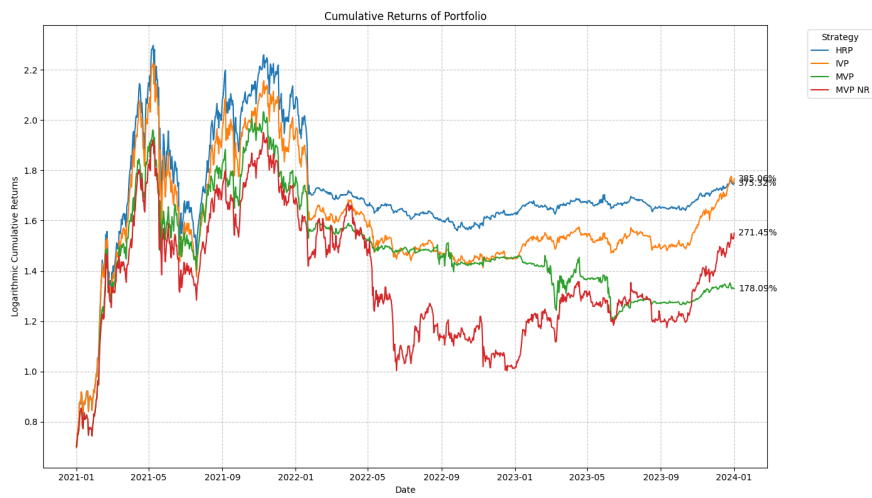


Figure 3: Cumulative Returns for Weekly Rebalancing.

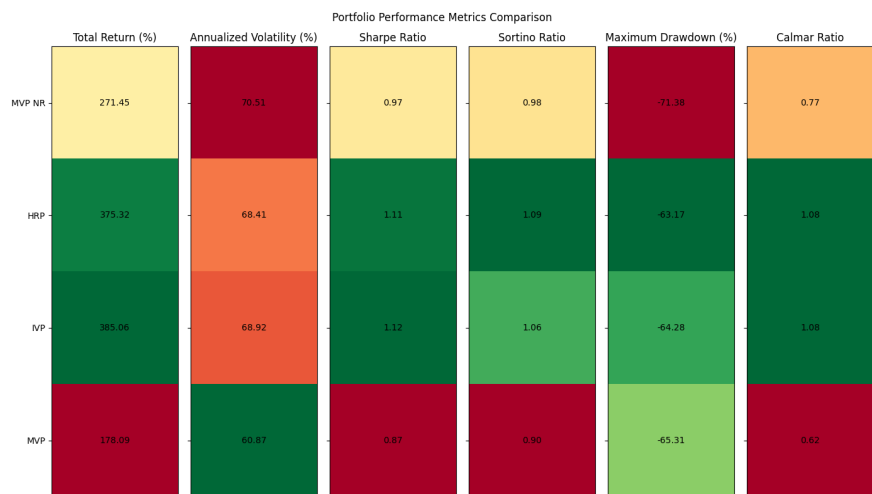


Figure 4: Metrics Evaluation for Weekly Rebalancing.